Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 8: Planar Graphs, the Mesh Matrix and Mesh Currents

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Colour code used in these notes and in the slides		
current i, meshes (1)	voltage v, e	nodes (1)

1. From nodes to meshes

The incidence matrix A gave two results earlier:

$$A\mathbf{i} = \mathbf{0} \quad \text{(KCL)}, \qquad \mathbf{v} = A^T\mathbf{e} \quad \text{(KVL)}.$$

With the $n-1$ node voltages as unknowns, KVL is satisfied automatically, and only KCL is left to write. This lecture builds the counterpart from the *meshes* of the graph: a matrix M that gives KVL as $M\mathbf{v} = \mathbf{0}$, and *mesh currents* for which KCL is satisfied automatically. It works for graphs that can be drawn on a plane without crossings.

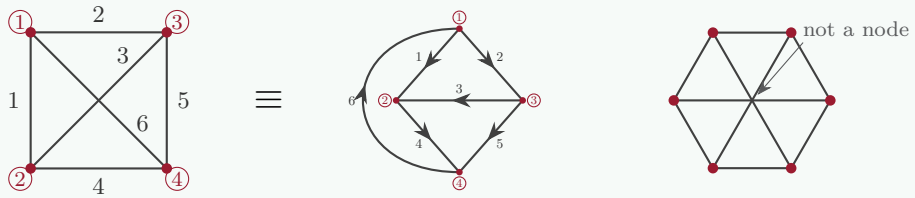
2. Planar graphs

Planar graph

A graph is **planar** if it can be drawn on a plane so that no two branches intersect at a point that is not a node.

The definition says *can be drawn*. A drawing with crossings does not make a graph nonplanar, if some other drawing has none. A planar circuit can be built on a one-layer board with no wire crossing another.

Example 8.1 — planar or not?



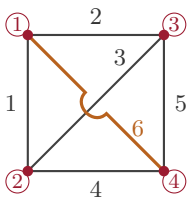
(a) branches 3 and 6 cross (b) the same graph, no crossing (c) nonplanar

(a) is the graph of the bridge circuit, drawn as a square: its two diagonals, branches 3 and 6, cross. (b) is the same graph — same nodes, same branches — with branch 6 taken around the outside. No branches cross, so the graph is planar.

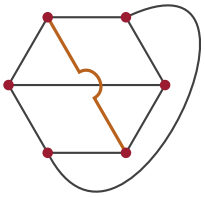
(c) has six nodes on a hexagon and three long diagonals that cross at the centre, which is not a node. However it is redrawn, two branches always cross: the graph is **nonplanar**.

2.1. Crossing is not touching

In drawing (a), branches 3 and 6 cross only on paper. They are two separate wires, and the crossing point is not a node. Circuit diagrams show this with a small **hop**: one wire jumps over the other. In real hardware the hop is a jumper wire, or a track on a second layer of the circuit board.



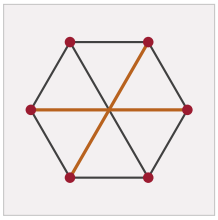
(a) with a hop: 6 jumps over 3



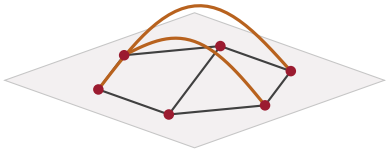
(c) the best possible: one hop

A graph is planar exactly when *some* drawing needs no hop. Drawing (a) needs one, the drawing (b) of the same graph needs none. The hexagon graph needs at least one hop however it is drawn.

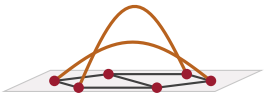
A hop is a step into the third dimension. Lift two diagonals of the hexagon graph to different heights and no two branches meet. Seen from above, the picture is drawing (c) again, with its crossings; from the side, the heights show.



from above



an oblique view

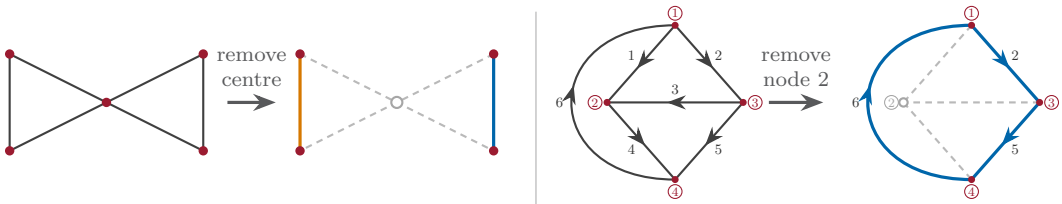


from the side

3. Nonseparable graphs

Nonseparable graph

A connected graph is **nonseparable** if it stays connected after removing any one node, together with all the branches attached to that node.



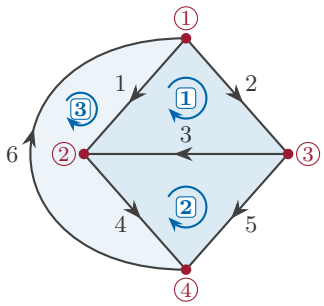
separable two pieces nonseparable (bridge) still one piece

Removing the centre node of the left graph removes its four branches and leaves two separate pieces, so that graph is separable. In the bridge graph, removing ② removes branches 1, 3 and 4; branches 2, 5, 6 still join ①, ③, ④. The same holds for any other node, so the bridge graph is nonseparable.

From now on the graph G is **planar and nonseparable**, and it is drawn without crossings.

4. Meshes

A planar drawing divides the plane into regions. Each region inside the graph is bounded by a loop of branches and contains no branch inside it. That loop is an **inner mesh** (or simply a mesh). The region outside the graph is not counted.



The bridge graph has three inner meshes:

- mesh 1: branches 1, 2, 3;
- mesh 2: branches 3, 4, 5;
- mesh 3: branches 1, 4, 6.

Each mesh gets a direction. By convention it is **clockwise**.

The number of inner meshes is

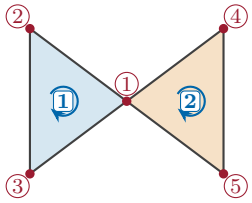
$$\ell = b - n + 1 \qquad (\text{here } 6 - 4 + 1 = 3),$$

the same as the number of independent KVL equations found earlier. A mesh is a loop, so each mesh gives one KVL equation, and these ℓ equations are independent.

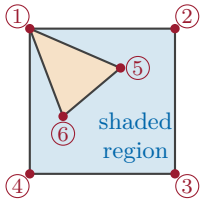
4.1. Why planar and nonseparable?

Planar. A mesh is a region of a drawing with no crossings. A nonplanar graph cannot be drawn without crossings, so it has no meshes. Its KVL equations are found another way, with trees, later on.

Nonseparable. This is a safe, conservative assumption: sufficient, not strictly necessary. It guarantees that every mesh is bounded by a single loop. A separable graph may or may not satisfy this, as the two examples below show.



(a) separable, each mesh is one loop



(b) separable, shaded region is not one loop

- **(a) Two triangles sharing 1.** Removing 1 splits the graph, so it is separable. Still, each mesh is a triangle, a single loop. The meshes work directly: $\ell = 2 = b - n + 1 = 6 - 5 + 1$.
- **(b) A triangle inside a square, attached at 1.** Again 1 is a **cut node**. The shaded region is bounded by the square *and* the triangle: two loops that touch only at 1. It is not a single loop, so it is not a proper mesh.

So the nonseparable assumption rules out case (b). Graphs like (b) can still be handled — there is a simple fix, which we will see in a coming lecture.

5. The mesh matrix

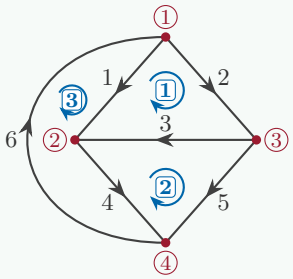
Mesh matrix

For a planar, nonseparable graph with ℓ inner meshes and b branches, the **mesh matrix** M is the $\ell \times b$ matrix with entries

$$m_{jk} = \begin{cases} +1 & \text{if branch } k \text{ is in mesh } j \text{ and their directions coincide,} \\ -1 & \text{if branch } k \text{ is in mesh } j \text{ and their directions are opposite,} \\ 0 & \text{otherwise.} \end{cases}$$

A row is a mesh: walk around it clockwise, and record each branch you meet with $+1$ if its arrow points the way you walk, -1 if it points against you.

Example 8.2 — the mesh matrix of the bridge



Mesh 1 (clockwise: ① → ③ → ② → ①). Branches 2 and 3 point along the walk: +1. Branch 1 points from ① to ②, against the walk: −1.

Mesh 2 (② → ③ → ④ → ②). Branch 5: +1. Branches 3 and 4: −1.

Mesh 3 (④ → ① → ② → ④, along branch 6 first). Branches 6, 1 and 4 all point along the walk: +1.

$$M = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{bmatrix} \end{array} \qquad (3 \times 6)$$

KVL in matrix form, with meshes

With the branch voltage vector $\mathbf{v} = [\,v_1 \ \cdots \ v_b\,]^T$, KVL around every mesh is

$$\boxed{M\mathbf{v} = \mathbf{0}}$$

For the bridge the three rows are

$$-v_1 + v_2 + v_3 = 0, \qquad -v_3 - v_4 + v_5 = 0, \qquad v_1 + v_4 + v_6 = 0.$$

Each is KVL around one mesh: going clockwise, a branch voltage counts with + when the walk goes from its + to its − (along the arrow), and with − otherwise.

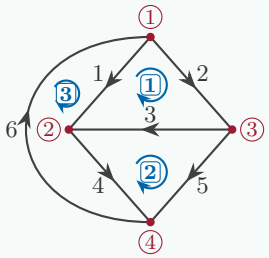
6. Mesh currents

Mesh current

To each inner mesh assign a **mesh current**: a current that circulates around that mesh in its (clockwise) direction. The mesh current vector is $\mathbf{i}_M = [\,i_{M1} \ \cdots \ i_{M\ell}\,]^T$.

Every branch current is a sum of mesh currents. A branch carries the mesh currents of the meshes it belongs to: + if the mesh direction coincides with the branch arrow, − if not. These are exactly the entries of column j of M.

Example 8.3 — branch currents of the bridge from mesh currents



$$\left.\begin{aligned} i_1 &= -i_{M1} + i_{M3} \\ i_2 &= i_{M1} \\ i_3 &= i_{M1} - i_{M2} \\ i_4 &= -i_{M2} + i_{M3} \\ i_5 &= i_{M2} \\ i_6 &= i_{M3} \end{aligned}\right\} \Longrightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \\ i_6 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{M^T} \begin{bmatrix} i_{M1} \\ i_{M2} \\ i_{M3} \end{bmatrix}$$

Branch 3 is shared by meshes 1 and 2. Mesh 1 goes along its arrow and mesh 2 against it, so $i_3 = i_{M1} - i_{M2}$. Branch 2 is only in mesh 1, so $i_2 = i_{M1}$.

Branch currents from mesh currents

$\boxed{\mathbf{i} = M^T \mathbf{i}_M}$ $\quad \mathbf{i} \in \mathbb{R}^b$: branch current vector, $\quad \mathbf{i}_M \in \mathbb{R}^\ell$: mesh current vector.

Remarks.

- Once the ℓ mesh currents are known, all b branch currents are known.
- KCL puts **no constraint** on the mesh currents. A mesh current enters each node on its loop and leaves it again, so it adds zero to KCL at every node. For example, KCL at ② of the bridge: $-i_1 - i_3 + i_4 = (i_{M1} - i_{M3}) - (i_{M1} - i_{M2}) + (-i_{M2} + i_{M3}) = 0$.
- In matrix form: $\mathbf{0} = A\mathbf{i} = AM^T\mathbf{i}_M$ for every $\mathbf{i}_M$. A matrix that gives zero for every vector is the zero matrix, so $AM^T = \mathbf{0}$. Write the rows of A and M as vectors,

$$A = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \mathbf{a}_3^T \end{bmatrix}, \qquad M = \begin{bmatrix} \mathbf{m}_1^T \\ \mathbf{m}_2^T \\ \mathbf{m}_3^T \end{bmatrix}, \qquad (AM^T)_{jr} = \mathbf{a}_j^T \mathbf{m}_r = 0 \quad \text{for all } j, r.$$

  Every row of A is **orthogonal** to every row of M. For example,

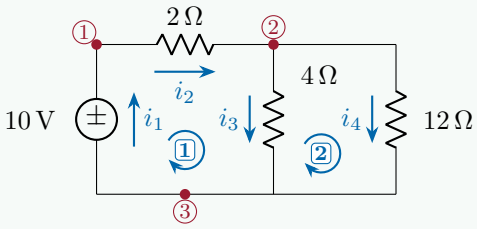
$$\mathbf{a}_2^T \mathbf{m}_1 = \begin{bmatrix} -1 & 0 & -1 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}^T = (-1)(-1) + (-1)(1) = 0.$$

  A node and a mesh share either no branch or two, and the two terms cancel.

So the mesh currents are a good choice of unknowns: there are only $\ell = b - n + 1$ of them, they are free, and KCL is then satisfied automatically. This is the idea behind mesh analysis.

Example 8.4 — mesh currents in a familiar circuit

The circuit below was solved earlier with node voltages: $i_1 = i_2 = 2\,\text{A}$, $i_3 = 1.5\,\text{A}$, $i_4 = 0.5\,\text{A}$, and $\mathbf{v} = [-10 \;\; 4 \;\; 6 \;\; 6]^T\,\text{V}$. Write M, express the branch currents with mesh currents, and check both results.



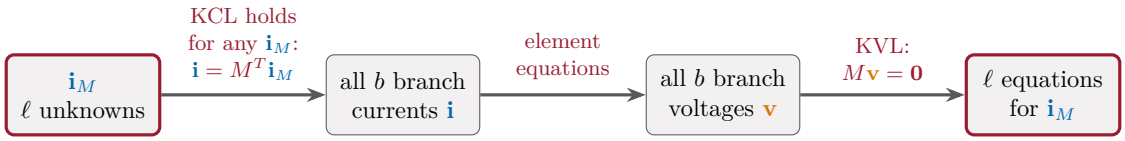
Mesh ① (left, clockwise) goes up through the source, right through $2\,\Omega$ and down through $4\,\Omega$: branches 1, 2, 3 all along their arrows. Mesh ② goes up through the $4\,\Omega$ (against branch 3) and down through the $12\,\Omega$ (along branch 4):

$$M = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \qquad \mathbf{i} = M^T\mathbf{i}_M : \quad i_1 = i_2 = i_{M1}, \quad i_3 = i_{M1} - i_{M2}, \quad i_4 = i_{M2}.$$

With $i_{M1} = 2\,\text{A}$ and $i_{M2} = 0.5\,\text{A}$ this gives exactly the branch currents above. KVL: $M\mathbf{v} = [-10 + 4 + 6, \quad -6 + 6]^T = \mathbf{0}$.

7. Looking ahead: why mesh currents?

The same idea as with node voltages, with the roles of KCL and KVL swapped. Again there are $2b$ unknowns and $2b$ equations: b element, $n - 1$ KCL and $\ell = b - n + 1$ KVL.



- **KCL.** Take the ℓ mesh currents as the unknowns and write every branch current as $\mathbf{i} = M^T\mathbf{i}_M$. This satisfies KCL at every node for *any* $\mathbf{i}_M$. So no KCL equation has to be written: the $n-1$ KCL equations hold automatically.
- **Element equations.** These are still used: they give each branch voltage from its current, hence in terms of $\mathbf{i}_M$. For a resistor shared by meshes 1 and 2, e.g. $v_k = R_k i_k = R_k(i_{M1} - i_{M2})$.
- **KVL.** Put these voltages into $M\mathbf{v} = \mathbf{0}$. This leaves ℓ equations, exactly as many as the unknowns $\mathbf{i}_M$.

Solve them for $\mathbf{i}_M$, and every other current and voltage follows. So $2b$ unknowns become $\ell = b - n + 1$, and $\ell < b$ because a circuit has $n \geq 2$ nodes. Node voltages need $n-1$ unknowns, mesh currents ℓ: which choice is shorter depends on the circuit. In Example 8.4, $\ell = 2$ and $n - 1 = 2$, but the source fixes one node voltage, so a single node voltage is enough there. This is **mesh analysis**, the subject of a coming lecture.

Summary

	node voltages	**mesh currents**
works for	any connected graph	planar, nonseparable graphs
how many unknowns	$n-1$	$\ell = b - n + 1$
matrix	A $((n-1)\times b)$	M $(\ell \times b)$
satisfied automatically	KVL: $\mathbf{v} = A^T\mathbf{e}$	KCL: $\mathbf{i} = M^T\mathbf{i}_M$
left to write	KCL: $A\mathbf{i} = \mathbf{0}$	KVL: $M\mathbf{v} = \mathbf{0}$

- A graph is **planar** if it can be drawn with no crossings, and **nonseparable** if removing any one node leaves it connected.
- The inner meshes of a planar drawing, taken clockwise, give the **mesh matrix** M: $+1$ along, -1 against, 0 otherwise. KVL is $M\mathbf{v} = \mathbf{0}$.
- **Mesh currents** circulate around the meshes; $\mathbf{i} = M^T\mathbf{i}_M$. KCL puts no constraint on them, so $AM^T = \mathbf{0}$.
- **Why**: with the ℓ mesh currents as unknowns, KCL holds automatically (no KCL equation to write), and KVL gives just enough equations — the idea of mesh analysis.

Next lecture

Nonplanar circuits: **trees**, **cutsets** and fundamental cutsets — tools that work for any connected graph.

Appendix: why the rows of A and M are orthogonal

Previously, $AM^T = \mathbf{0}$ came from KCL: $\mathbf{i} = M^T \mathbf{i}_M$ satisfies $A\mathbf{i} = \mathbf{0}$ for every $\mathbf{i}_M$. Here we see the same fact directly from the entries of A and M.

The entries. Recall

$$a_{jk} = \begin{cases} +1 & \text{branch } k \text{ leaves node } j \\ -1 & \text{branch } k \text{ enters node } j \\ 0 & \text{otherwise} \end{cases} \qquad m_{rk} = \begin{cases} +1 & \text{branch } k \text{ on mesh } r, \text{ along its direction} \\ -1 & \text{branch } k \text{ on mesh } r, \text{ against it} \\ 0 & \text{otherwise} \end{cases}$$

and $\mathbf{a}_j^T \mathbf{m}_r = \sum_k a_{jk}\, m_{rk}$. A term is nonzero only if branch k touches node j *and* lies on mesh r.

Step 1 — which branches count. Walk once around mesh r in its direction.

- If the walk never reaches node j, no branch is shared: $\mathbf{a}_j^T \mathbf{m}_r = 0$.
- If it does, it comes into node j on one branch and goes out on another. A mesh is a loop, so it passes node j only once. Exactly two terms are nonzero.

Step 2 — the sign of each term. Both signs depend on the branch arrow: a_{jk} compares it with node j, m_{rk} with the walk. Reversing the arrow flips both signs, so their product stays the same. The product depends only on the walk:

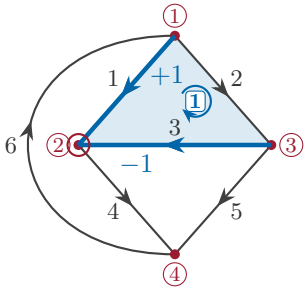
	branch arrow along the walk	**branch arrow against the walk**
walk goes out of j	 $a_{jk} = +1,\ m_{rk} = +1$ $(+1)(+1) = +1$	 $a_{jk} = -1,\ m_{rk} = -1$ $(-1)(-1) = +1$
walk comes into j	 $a_{jk} = -1,\ m_{rk} = +1$ $(-1)(+1) = -1$	 $a_{jk} = +1,\ m_{rk} = -1$ $(+1)(-1) = -1$

blue: the walk around mesh r; black: branch k and its arrow

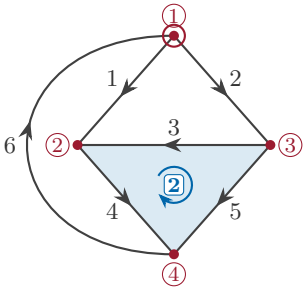
The branch the walk goes out on always gives $+1$; the branch it comes in on always gives -1.

Step 3 — the sum. $\mathbf{a}_j^T \mathbf{m}_r = (+1) + (-1) = 0$.

Example: the bridge.



node ② and mesh ①: out on 1 (+1), in on 3 (−1)



node ① and mesh ②: no shared branch

- **Node ②, mesh ①.** The walk ① → ③ → ② → ① comes into ② on branch 3 and goes out on branch 1. Branch 3: $a = -1$ (enters ②), $m = +1$ (along), product -1. Branch 1: $a = -1$ (enters ②), $m = -1$ (against), product $+1$. The sum is 0, as computed earlier.
- **Node ①, mesh ②.** Mesh ② does not pass ①, so no branch is shared and the product is 0.

What it means. A single mesh current flows into node j on one branch and out on the other, so it adds nothing to KCL at node j. Steps 1–3 are this sentence written with the signs of A and M.

Remarks. Only the closed walk was used, so the result holds for any loop, not only for meshes. Reversing the direction of a mesh flips the sign of its whole row; the product is still zero.